

BTS Reporting Carrier On-Time Performance

Raw Data Field Reference — Source Documentation for the BTS_Transformation Data Warehouse

This document describes the raw source table behind the **RAW_FLIGHTS_*** tables in **BTS_AIRLINE_DB**. It is the field-level reference for the flight-level dataset downloaded from the BTS TranStats Download Center, table **On-Time : Reporting Carrier On-Time Performance** (coverage 1987-present, monthly release, one row per non-stop domestic flight segment).

Source

Bureau of Transportation Statistics (BTS), TranStats Download Center — transtats.bts.gov/DL_SelectFields.aspx, table ID ONTIME_REPORTING. Data is published monthly by the US DOT, with the raw source available as CSV inside a per-month ZIP file.

Grain

One row per reported non-stop domestic flight segment operated by a US carrier, per day. This is the level at which the pipeline ingests data into RAW_FLIGHTS_2024, RAW_FLIGHTS_2025, and RAW_FLIGHTS_2026, which are unioned by `stg_flights` in the dbt staging layer.

How to read the tables below

Fields are grouped exactly as they appear on the BTS field-selection page.

Time Period

Field Name	Description
Year	Year
Quarter	Quarter (1-4)
Month	Month
DayofMonth	Day of Month
DayOfWeek	Day of Week
FlightDate	Flight Date (yyyymmdd)

Airline

Field Name	Description
Reporting_Airline	Unique Carrier Code. When the same code has been used by multiple carriers, a numeric suffix is used for earlier users (e.g. PA, PA(1), PA(2)). Use this field for analysis across a range of years.
DOT_ID_Reporting_Airline	ID assigned by US DOT to identify a unique airline (carrier), defined as one holding and reporting under the same DOT certificate regardless of Code, Name, or holding company.
IATA_CODE_Reporting_Airline	Code assigned by IATA to identify a carrier. The same code may have been assigned to different carriers over time, so it is not always unique — use the Unique Carrier Code for analysis.
Tail_Number	Tail Number
Flight_Number_Reporting_Airline	Flight Number

Origin

Field Name	Description
OriginAirportID	Origin Airport, Airport ID. Assigned by US DOT to identify a unique airport. Use this field for airport analysis across years since airport codes can change and be reused.
OriginAirportSeqID	Origin Airport, Airport Sequence ID. Identifies a unique airport at a given point in time; airport attributes such as name or coordinates may change over time.
OriginCityMarketID	Origin Airport, City Market ID. Identifies a city market; used to consolidate airports serving the same city market.
Origin	Origin Airport (code)
OriginCityName	Origin Airport, City Name
OriginState	Origin Airport, State Code
OriginStateFips	Origin Airport, State FIPS
OriginStateName	Origin Airport, State Name

Field Name	Description
OriginWac	Origin Airport, World Area Code

Destination

Field Name	Description
DestAirportID	Destination Airport, Airport ID. Assigned by US DOT to identify a unique airport. Use this field for airport analysis across years since airport codes can change and be reused.
DestAirportSeqID	Destination Airport, Airport Sequence ID. Identifies a unique airport at a given point in time; airport attributes such as name or coordinates may change over time.
DestCityMarketID	Destination Airport, City Market ID. Identifies a city market; used to consolidate airports serving the same city market.
Dest	Destination Airport (code)
DestCityName	Destination Airport, City Name
DestState	Destination Airport, State Code
DestStateFips	Destination Airport, State FIPS
DestStateName	Destination Airport, State Name
DestWac	Destination Airport, World Area Code

Departure Performance

Field Name	Description
CRSDepTime	Scheduled (CRS) Departure Time, local time HHMM
DepTime	Actual Departure Time, local time HHMM
DepDelay	Difference in minutes between scheduled and actual departure time. Early departures show negative numbers.
DepDelayMinutes	Difference in minutes between scheduled and actual departure time. Early departures set to 0.
DepDel15	Departure Delay Indicator, 15 Minutes or More (1 = Yes)
DepartureDelayGroups	Departure Delay intervals, every 15 minutes from <-15 to >180
DepTimeBlk	Scheduled Departure Time Block, Hourly Intervals
TaxiOut	Taxi Out Time, in Minutes
WheelsOff	Wheels Off Time, local time HHMM

Arrival Performance

Field Name	Description
WheelsOn	Wheels On Time, local time HHMM
TaxiIn	Taxi In Time, in Minutes
CRSArrTime	Scheduled (CRS) Arrival Time, local time HHMM
ArrTime	Actual Arrival Time, local time HHMM
ArrDelay	Difference in minutes between scheduled and actual arrival time. Early arrivals show negative numbers.
ArrDelayMinutes	Difference in minutes between scheduled and actual arrival time. Early arrivals set to 0.
ArrDel15	Arrival Delay Indicator, 15 Minutes or More (1 = Yes)
ArrivalDelayGroups	Arrival Delay intervals, every 15 minutes from <-15 to >180
ArrTimeBlk	Scheduled Arrival Time Block, Hourly Intervals

Cancellations and Diversions

Field Name	Description
Cancelled	Cancelled Flight Indicator (1 = Yes)
CancellationCode	Specifies the Reason for Cancellation
Diverted	Diverted Flight Indicator (1 = Yes)

Flight Summaries

Field Name	Description
CRSElapsedTime	Scheduled (CRS) Elapsed Time of Flight, in Minutes
ActualElapsedTime	Elapsed Time of Flight, in Minutes
AirTime	Flight Time, in Minutes
Flights	Number of Flights
Distance	Distance between airports, in Miles
DistanceGroup	Distance Intervals, every 250 Miles, for Flight Segment

Cause of Delay (data starts 6/2003)

Field Name	Description
CarrierDelay	Carrier Delay, in Minutes
WeatherDelay	Weather Delay, in Minutes
NASDelay	National Air System Delay, in Minutes
SecurityDelay	Security Delay, in Minutes

Field Name	Description
LateAircraftDelay	Late Aircraft Delay, in Minutes

Gate Return Information at Origin Airport (data starts 10/2008)

Field Name	Description
FirstDepTime	First Gate Departure Time at Origin Airport
TotalAddGTime	Total Ground Time Away from Gate for Gate Return or Cancelled Flight
LongestAddGTime	Longest Time Away from Gate for Gate Return or Cancelled Flight

Diverted Airport Information (data starts 10/2008)

Field Name	Description
DivAirportLandings	Number of Diverted Airport Landings
DivReachedDest	Diverted Flight Reaching Scheduled Destination Indicator (1 = Yes)
DivActualElapsedTime	Elapsed Time of Diverted Flight Reaching Scheduled Destination, in Minutes. ActualElapsedTime remains NULL for all diverted flights.
DivArrDelay	Difference in minutes between scheduled and actual arrival time for a diverted flight reaching scheduled destination. ArrDelay remains NULL for all diverted flights.
DivDistance	Distance between scheduled destination and final diverted airport, in Miles. Value is 0 for a diverted flight reaching scheduled destination.
Div1Airport ... Div5Airport	Diverted Airport Code (repeated for up to 5 diversions)
Div1AirportID ... Div5AirportID	Airport ID of Diverted Airport (Unique Key for an Airport)
Div1AirportSeqID ... Div5AirportSeqID	Airport Sequence ID of Diverted Airport (Unique Key for Time-Specific Airport Info)
Div1WheelsOn ... Div5WheelsOn	Wheels On Time at Diverted Airport, local time HHMM
Div1TotalGTime ... Div5TotalGTime	Total Ground Time Away from Gate at Diverted Airport
Div1LongestGTime ... Div5LongestGTime	Longest Ground Time Away from Gate at Diverted Airport
Div1WheelsOff ... Div5WheelsOff	Wheels Off Time at Diverted Airport, local time HHMM
Div1TailNum ... Div5TailNum	Aircraft Tail Number for Diverted Airport

Notes for BTS_Transformation Ingestion

The points below reflect how this raw structure is actually handled in the pipeline, for reference alongside the field list.

- BTS represents nulls as empty strings rather than true NULLs — every numeric raw column is read with TRY_CAST in the staging layer to avoid load failures.
- All HHMM time fields (CRSDepTime, DepTime, WheelsOff, WheelsOn, CRSArrTime, ArrTime) are kept as VARCHAR; duration/minutes columns are cast to INTEGER.
- Cancelled flights can legitimately carry a DepDelay (delayed before being cancelled) but never an ArrDelay, since the flight never arrived.
- CancellationCode is decoded inline into a Cancellation_Reason varchar in the mart layer — there is no separate dim_cancellation table.
- Origin/Dest and their related airport attribute columns both resolve to the single role-playing dim_airport dimension rather than separate origin/destination dimensions.
- The Diverted Airport Information group (Div1... through Div5...) is sparsely populated — most rows have Diverted = 0 and all Div* fields empty.